

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B2. SYMMETRY AND RELATIVITY

TRINITY TERM 2023

Thursday, 8 June, 2:30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. In this question units are adopted such that $c = 1$.

(a) Prove that (i) a single photon cannot decay into an electron-positron pair, (ii) a single charged particle cannot emit a single photon by itself. [5]

(b) In the laboratory frame, a proton (p) of mass m and 4-momentum $(E_p, \vec{p}_p)$ collides with a stationary anti-proton ($\bar{p}$). Show that in order to thus form a particle of mass M the required total energy in the laboratory frame grows as M^2 . If the attainable energy for protons is limited would it be better to use colliding p- $\bar{p}$ beams? How would the threshold energy then depend on M ? [5]

(c) A pion (π^+) at rest decays into a muon neutrino (ν_μ) and an anti-muon (μ^+) via the reaction $\pi^+ \rightarrow \mu^+ + \nu_\mu$. Taking the mass of the neutrino to be zero, (i) show that the kinetic energy of the anti-muon is $(m_\pi - m_\mu)^2/(2m_\pi)$ and (ii) calculate the decay length of the anti-muon in the π^+ rest frame given that its proper lifetime is τ_μ , giving your answer exclusively in terms of τ_μ and the masses in the problem. [5]

(d) The rest mass of any bound atomic nucleus is less than the sum of the masses of its constituents, owing to binding energy. Let m_n , m_p and m_μ be the rest masses of a free neutron, proton and muon, respectively. A neutron is at rest in a nucleus and the energy required to eject it (the binding energy) is E_b . An incoming neutrino collides with this neutron, such that the outcome is a proton and a muon, both of which are ejected with final energies E_p , E_μ respectively. Prove that

$$E_\nu = \frac{m_p^2 - (m_n - E_b)^2 - m_\mu^2 + 2E_\mu(m_n - E_b)}{2(m_n - E_p - E_\mu + |\vec{p}_\mu| \cos \theta_\mu)},$$

where θ_μ is the angle of the muon's trajectory with respect to that of the neutrino. [7]

(e) We've assumed the neutrino has zero mass but the result will apply to good approximation if the neutrino mass is small but non-zero. It is observed that neutrinos in otherwise empty space change from one kind to another as time goes on. Explain why this observation implies that neutrinos have non-zero mass (answer in a few sentences—no particle physics knowledge is needed). [3]

2. (a) Two synchronised clocks A and B lie on the x -axis of a frame S at $x = 0$ and $x = L$ respectively. They are observed in a frame S' in standard configuration with S (that is, S' moves with respect to S in the x direction and the axes are aligned), with boost parameters γ and β . What is the time difference per tick between these clocks in frame S'? Which clock is ahead?

[3]

(b) A cylinder is initially at rest in frame S, with its axis of symmetry along the x axis. A straight line parallel to the x -axis is painted on the outer surface. Let ϕ be the angle between the y axis and a radius vector from the cylinder's axis to this painted line. The cylinder is then made to rotate about its axis with a rigid motion in frame S at constant angular velocity ω . (Rigid motion here means that all parts of the cylinder have this same angular velocity in frame S).

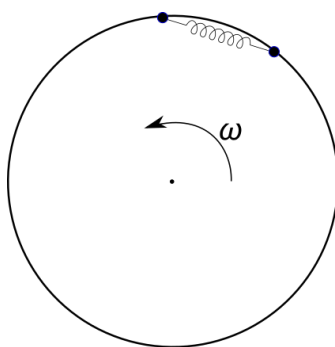
Observed in frame S', the line on the surface of the cylinder is not straight. In order to describe it quantitatively, let ϕ' be the angle between the y' axis and a radius vector from the cylinder's axis to the painted line in S'. Obtain the change in this angle ϕ' per unit length along the cylinder in S'. Hence describe the shape of the painted line in S'. Explain how these observations concerning the cylinder relate to the observations of the clocks in part (a).

[7]

(c) (i) A disc of radius R rotates with constant angular velocity ω . An experimenter measures the ratio of circumference and diameter of the disc using a set of infinitesimal sticks (of known and equal proper length) attached to the disk. The ratio is taken as the number of such sticks around the circumference to the number along a radius. What value will the observer obtain?

(ii) Now consider such a disc prepared initially at rest in an inertial frame S, with the ends of a spring fixed to two points on the circumference, such that the spring has its natural length and therefore exerts no force. The whole assembly is then set in rigid motion such that all parts have the same angular velocity ω in S, they acquire this angular velocity abruptly and simultaneously in S, and the radius of the disc is unchanged (see figure). Explain whether the spring is now in tension, in compression, or neither. How does this observation relate to the ratio discussed in part (i)?

[5]



(d) A particle moves in the y - z plane of frame S with energy E and 3-momentum $\vec{p}$. Using a suitable second rank tensor, or otherwise, find expressions for the components of the angular-momentum 3-vector $\vec{L}'$ as observed in frame S', in terms of E and components of $\vec{p}$.

[5]

(e) Show that $L^2 - (ct\vec{p} - E\vec{x}/c)^2$ is an invariant, where the symbols have their usual meanings ($\vec{L}$ is the angular momentum 3-vector).

[5]

3. (a) The electromagnetic field tensor is $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$ where A^μ is the 4-vector potential $(\phi/c, \vec{A})$. Show that the components of $F^{\mu\nu}$ are $-F^{0i} = F^{i0} = E_i/c$ and $F^{ij} = -\epsilon_{ijk}B_k$ where ϵ_{ijk} is the completely antisymmetric Levi-Cevita symbol. You may quote relationships between fields and potentials. [3]

(b) The rate of change of 4-momentum with respect to the proper time of a charged particle is given by $\frac{dp^\alpha}{d\tau} = qF^\alpha_\beta u^\beta$. Obtain from this the Lorentz-force law (in a general electromagnetic field), and an expression for the rate of change of the particle's energy in terms of the field and velocity. [3]

(c) The 4-potential due to a moving charged particle depends on the spacetime separation between the source event $(ct_s, \vec{x}_s)$ on the charge's worldline and the field event $(ct_f, \vec{x}_f)$ where the field is measured. A particle with charge q moves along the x axis of frame S with worldline $(x, y, z)_q = (vt, 0, 0)$. Use a suitable frame and a Lorentz transformation to show that the 4-potential in S at events on the x -axis has the form

$$\frac{q}{4\pi\epsilon_0 c\sigma} (1, \beta, 0, 0),$$

where your answer should give an expression for σ in terms of suitable quantities, making it very clear at which events the quantities are to be evaluated or to which events they refer. [6]

(d) The 4-potential of an arbitrarily moving point charge can be written

$$A^\alpha = \frac{q}{4\pi\epsilon_0(-R^\mu U_\mu)} \frac{U^\alpha}{c}.$$

Fully define the quantities U^α and R^μ here, giving your reasoning. Show that the same 4-potential is obtained as in part (c). [5]

(e) A charged particle's trajectory in the x - y plane is given by $(\rho \cos \omega t_s, \rho \sin \omega t_s, 0)$ where ρ and ω are constants. Find E_z (the z component of the electric field) on the z axis for $z \neq 0$. [8]

4. (a) Derive the velocity addition formula

$$w = \frac{u + v}{1 + uv/c^2}. \quad [3]$$

- (b) Show that the 4-acceleration a^μ of a particle is always orthogonal to its 4-velocity and if the particle is at rest then a^μ has only spatial components. [4]

- (c) A rocket of mass $M(\tau)$ and 4-velocity u^α propels itself by ejecting propellant at rate $dm/d\tau$ where dm is the mass of propellant ejected in proper time $d\tau$ along the rocket's worldline. Let v^α be the components of the 4-velocity of the ejected propellant as observed in the instantaneous rest-frame (IRF) of the rocket. For motion in one spatial dimension, consider the ejection of propellant mass dm and write down the condition for conservation of 4-momentum in terms of u^α , v^α , du^α , M , dM , dm at first order in small quantities. [Hint: keep in mind that conservation laws apply to energy and momentum not mass.] [4]

- (d) By considering your 4-vector expression in the IRF, or otherwise, show that

$$\frac{dM}{M} = -\frac{du}{v},$$

where du is the rocket's speed at $\tau + d\tau$ relative to the frame which was the IRF at τ , and v is the speed of the ejected propellant in that frame. [5]

- (e) Let w be the speed of the rocket relative to its launch site. Show that the expression for the rocket's mass M in terms of this speed takes the form

$$M(w) = M(0) \left(\frac{1 - \frac{w}{c}}{1 + \frac{w}{c}} \right)^\alpha,$$

where your answer should give the expression for the exponent α in terms of relevant quantities.

[You may need the integral $\int \sec \theta d\theta = \frac{1}{2} \ln \left| \frac{1 + \sin \theta}{1 - \sin \theta} \right| + \text{constant}.$] [9]